

Network Wars — battle model comparison

12,670 ground-truth live iOS battles · closed-form (shipped) vs best-fit hybrid loop

MODELS

Live data	12,670 single-battle observations (a attacker vs d defender), per-(a,d) capture rate + survivor counts.
Closed form (3 separate sub-models)	SHIPPED engine. Outcome is ONE Bernoulli draw: $P(\text{capture}) = a^{3.40} / (a^{3.40} + 1.26 d^{3.40})$. Survivors are SEPARATE fitted draws: occupier = 1 + BetaBinomial(a-2; mean 0.82a-0.44d+0.1, rho=0.21); remnant = Binomial(d; mean 0.30+0.24d+0.42*max(0,d-a)).
Hybrid loop (1 loop + 1 remnant rule)	A LETHAL-attacker play-out gives the WIN and the OCCUPIER; the repel REMNANT is a DECOUPLED draw (the play-out cannot make it -- depletion-repels self-select for low damage, and the real loss scales with d). Best model found by FULL joint likelihood (outcome + every survivor count): <pre> while a > 1 and d > 0: if rng() < 0.50: d -= 1 # attacker volley lands (lethal) if rng() < 0.28: a -= 1 # defender volley lands elif rng() < 0.05*d^0.53/(d^0.53+4.98*a^0.53): repelled = True; break # attacker routs if d==0: capture, occupier = a-1 (source keeps 1) else: repel, remnant = Binomial(d, mu/d), mu = 0.30+0.24d+0.42*max(0,d-a) </pre>

FIT SUMMARY (lower is better)

	outcome AIC	occupier-mean RMSE	remnant-mean RMSE
Closed form	10434	0.242	0.097
Hybrid loop	11013	0.224	0.097

(on the FULL joint likelihood the hybrid is the BEST of the three: +1729 nats over the noise floor, vs pure loop +1988 and closed form +2342 -- it pairs the loop's better occupier with the hinge remnant.)

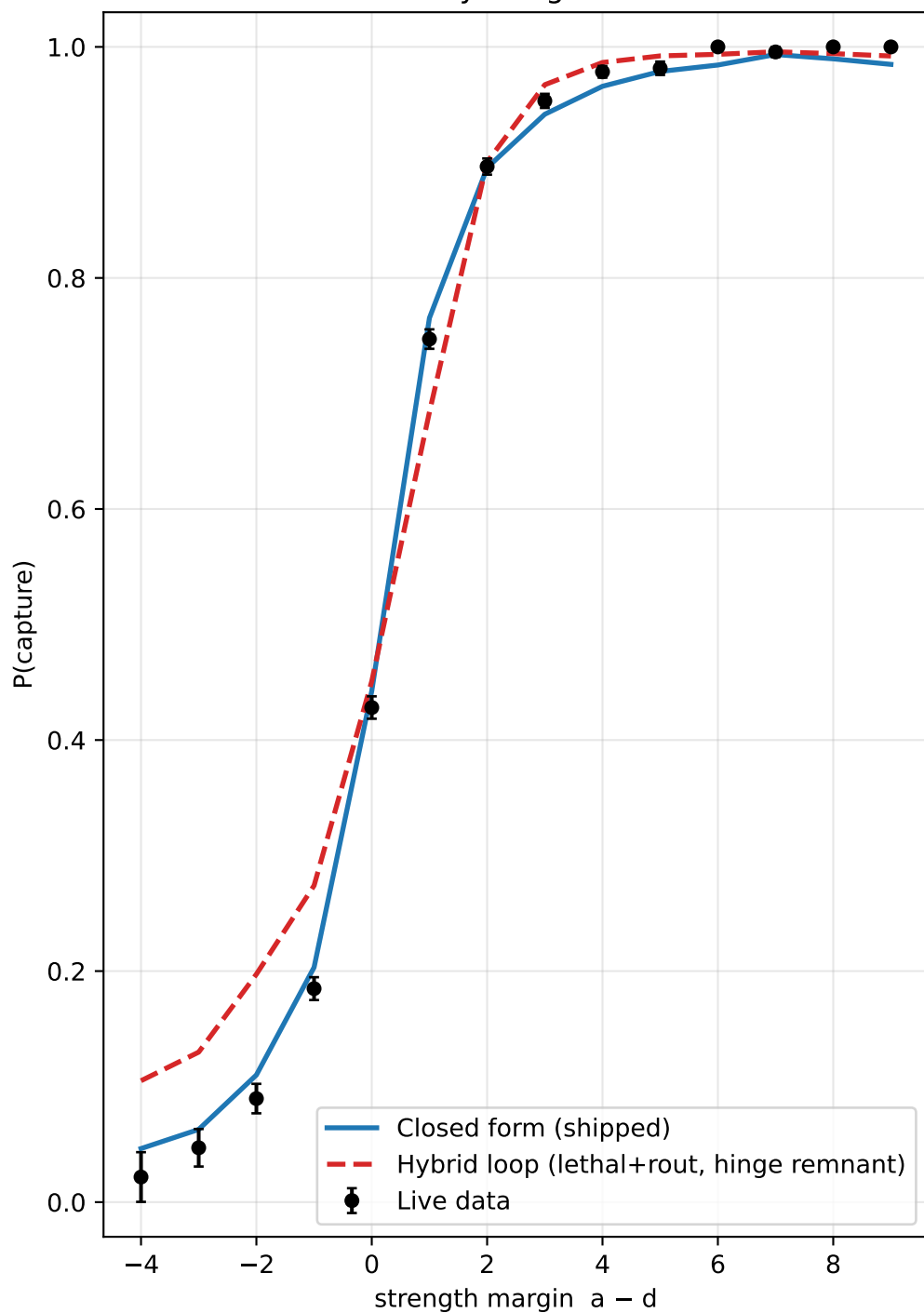
READING THE FIGURES

p2 who-wins by margin + residual · p3 ratio-sensitivity · p4 occupier survivors · p5 remnant survivors
p6 survivor DISTRIBUTIONS (spread, not just mean) · p7 per-cell calibration.

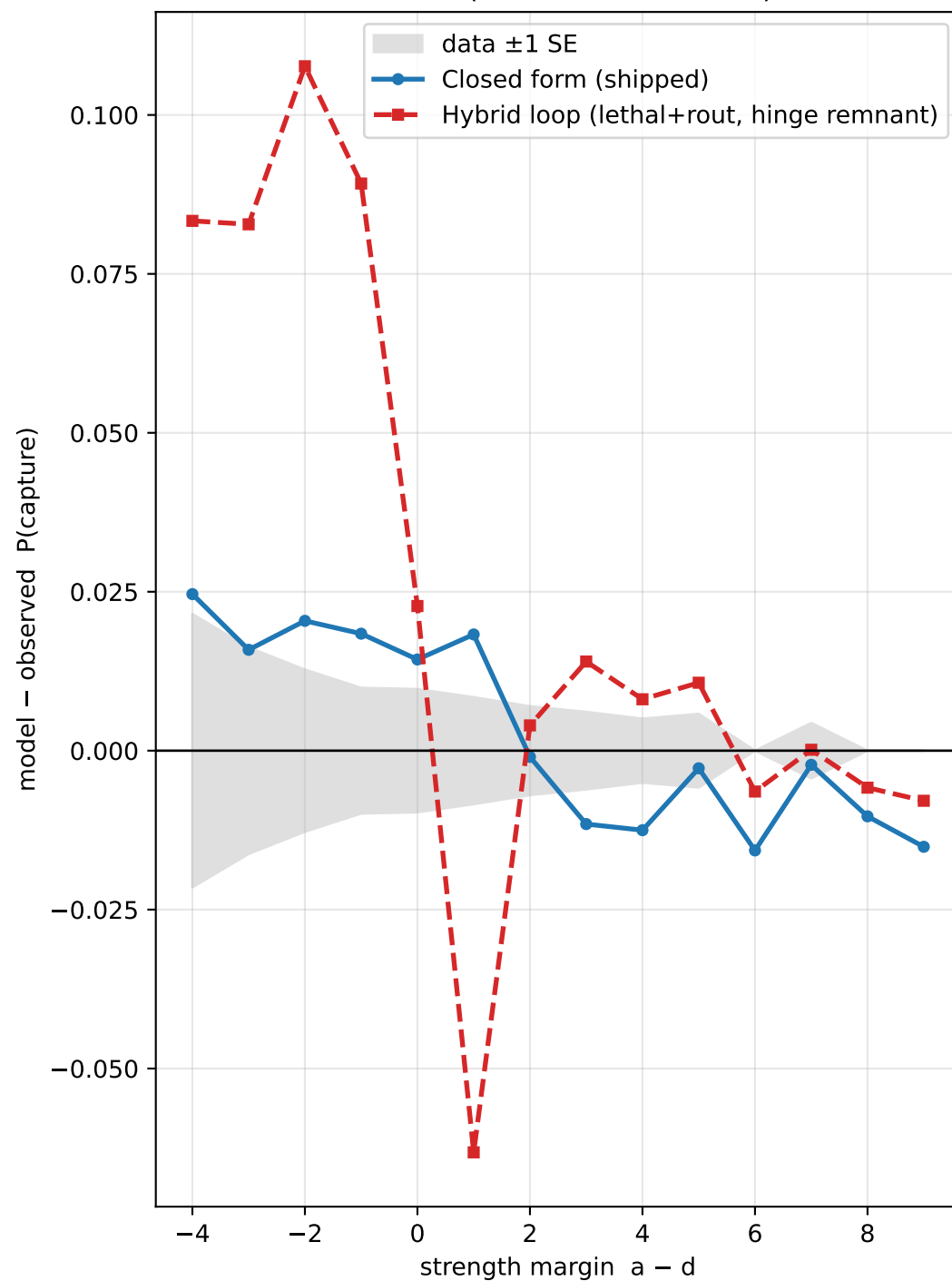
TAKEAWAY The play-out loop matches the occupier survivors BETTER than the closed form and tracks the win curve, but it structurally cannot produce the repel remnant: a depletion-repel is exactly the case where the attacker killed few defenders (conditioning), and the real loss scales with the defender's size d . So the remnant is given its own decoupled (a, d) draw -- the documented 'assault guts the defender' hinge, now shared with the closed form (p5/p6: the two curves coincide). One iterated mechanism for WIN+occupier, one line for the remnant; the result is the best-fitting model overall. Residual honesty: the loop's win curve is still a touch generous to underdogs (p2 right, lower-left) -- the price of one mechanism doing outcome and occupier together.

Who wins (capture probability)

by margin

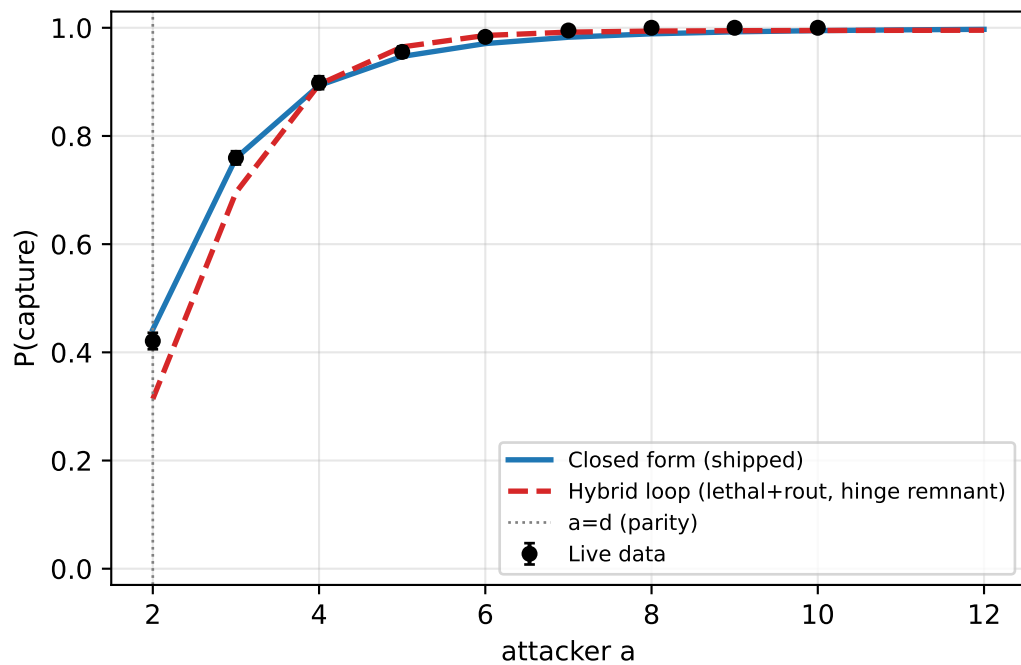


residual (closer to 0 = better)

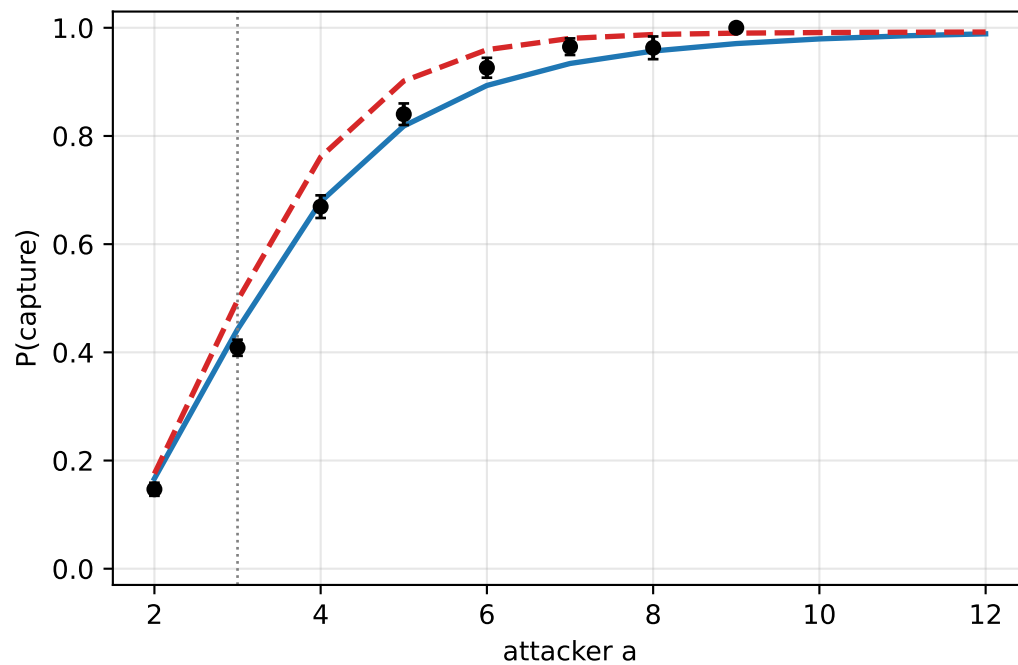


Ratio-sensitivity — capture vs attacker size at fixed defender size

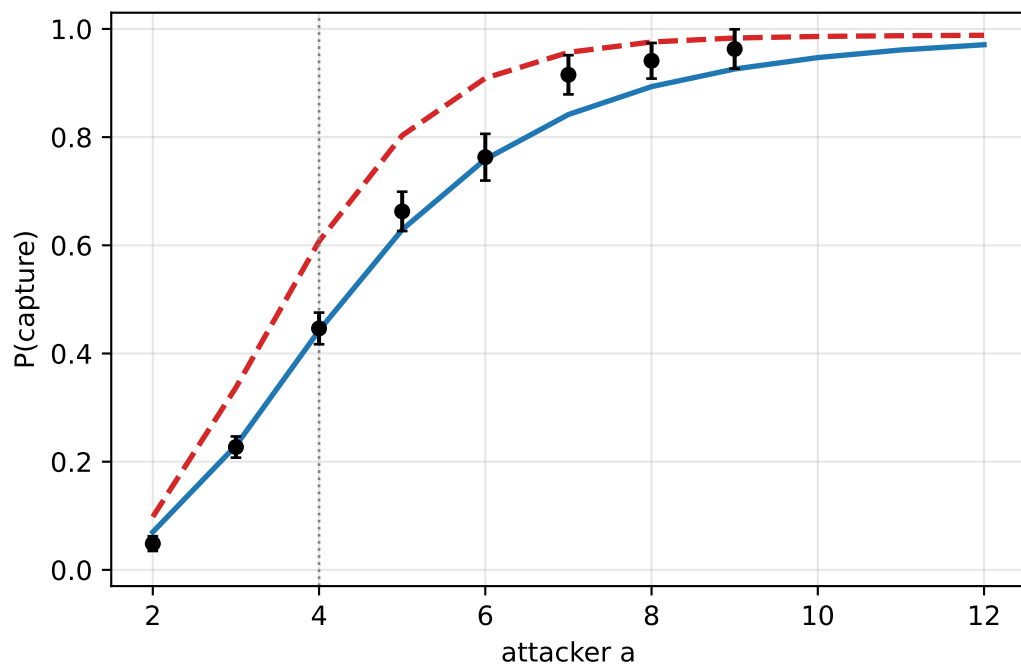
defender $d = 2$



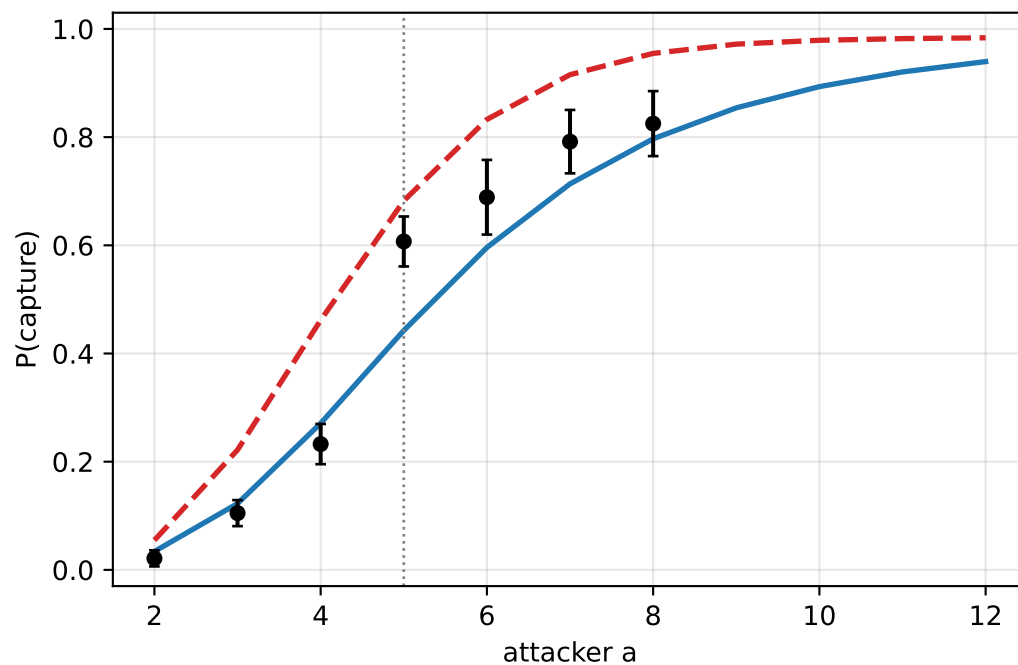
defender $d = 3$



defender $d = 4$

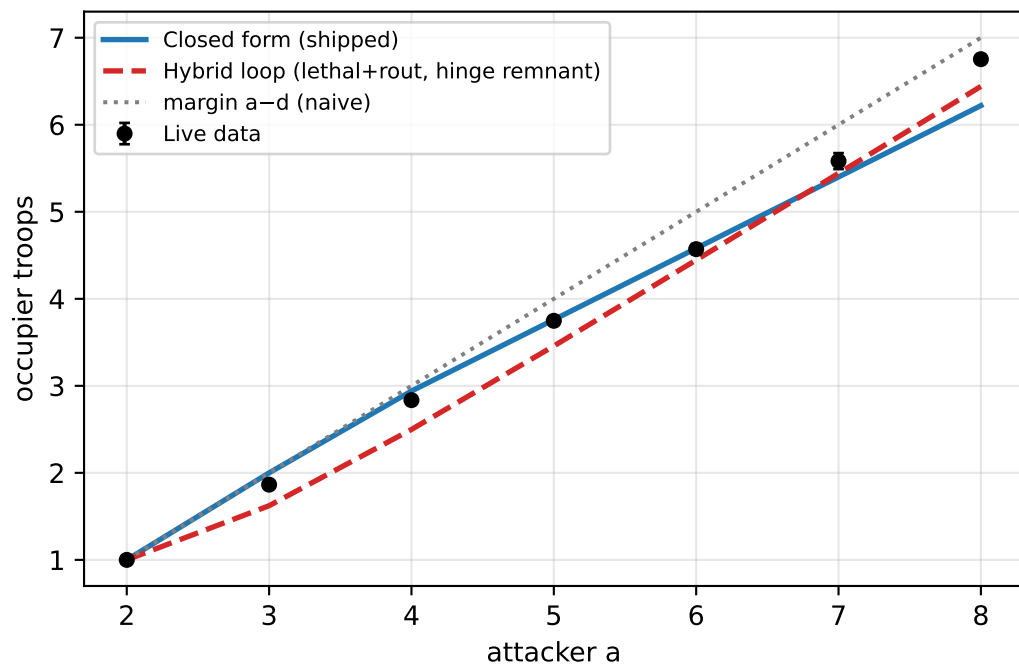


defender $d = 5$

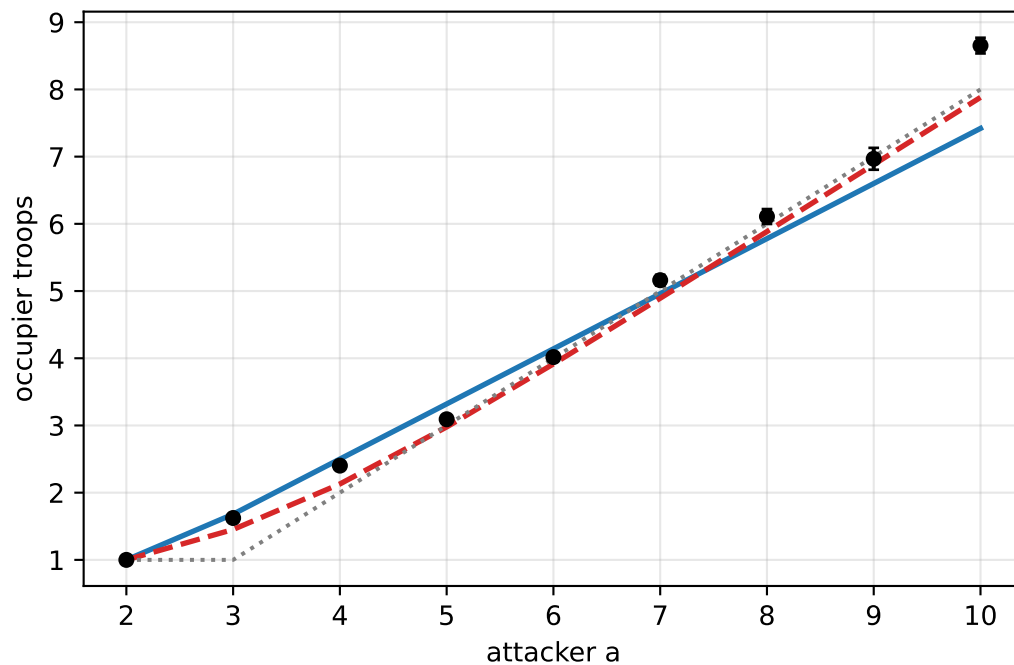


Occupier survivors after a CAPTURE — mean troops left on the taken node (vs a-d=margin alone)

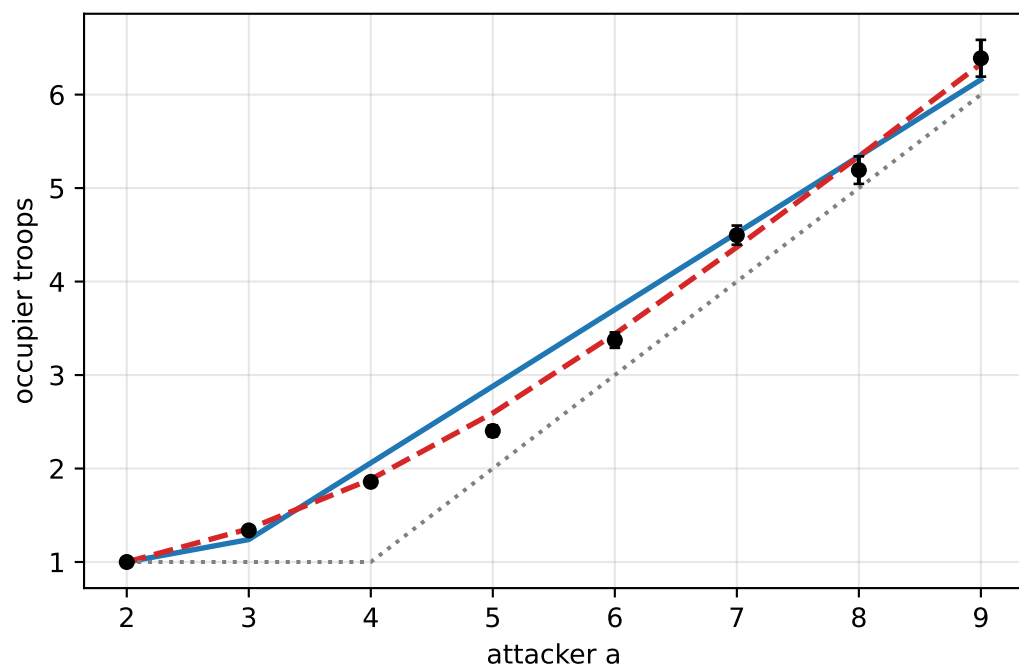
defender d = 1



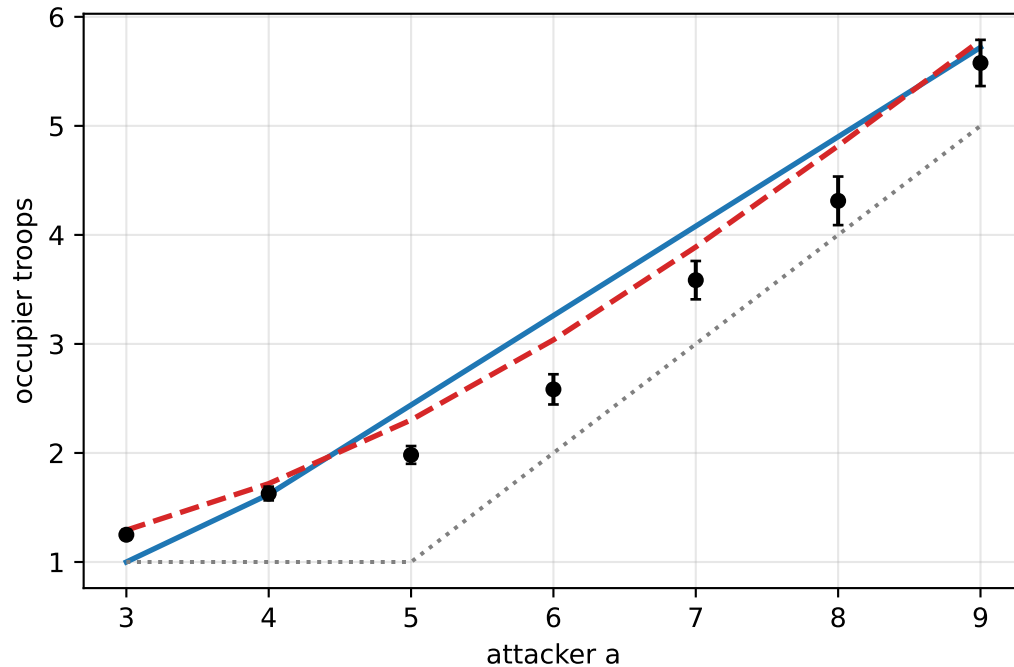
defender d = 2



defender d = 3

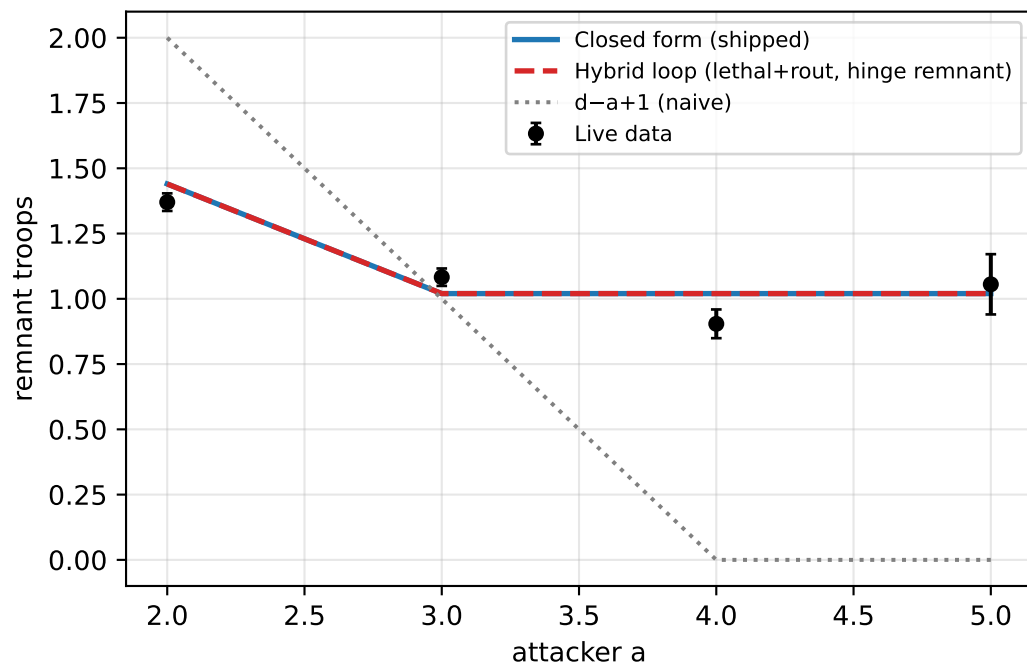


defender d = 4

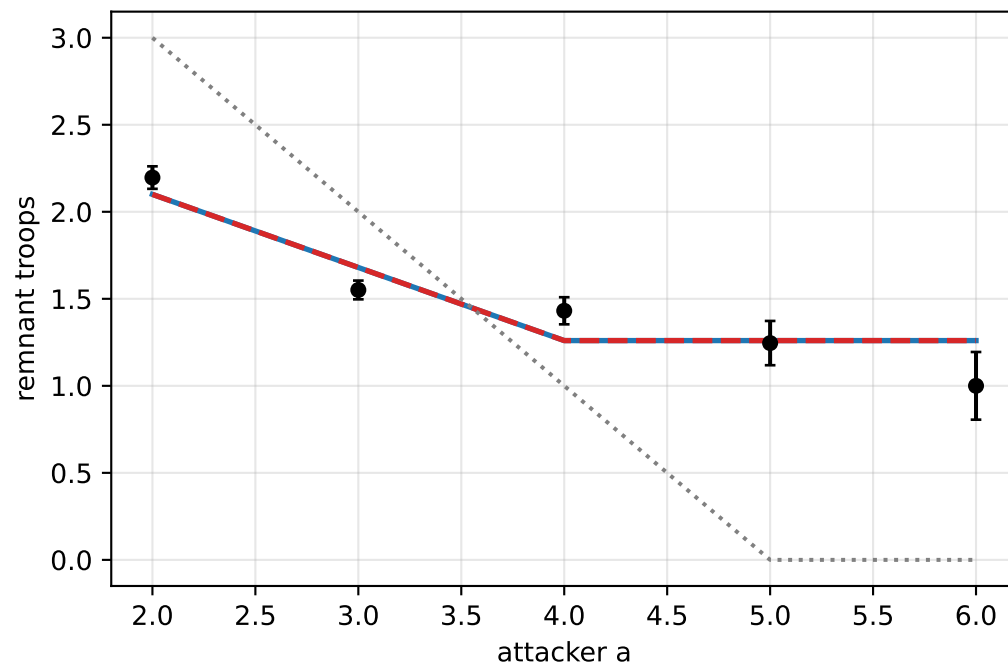


Defender remnant after a REPEL — mean troops left on the held node (assault guts the defender)

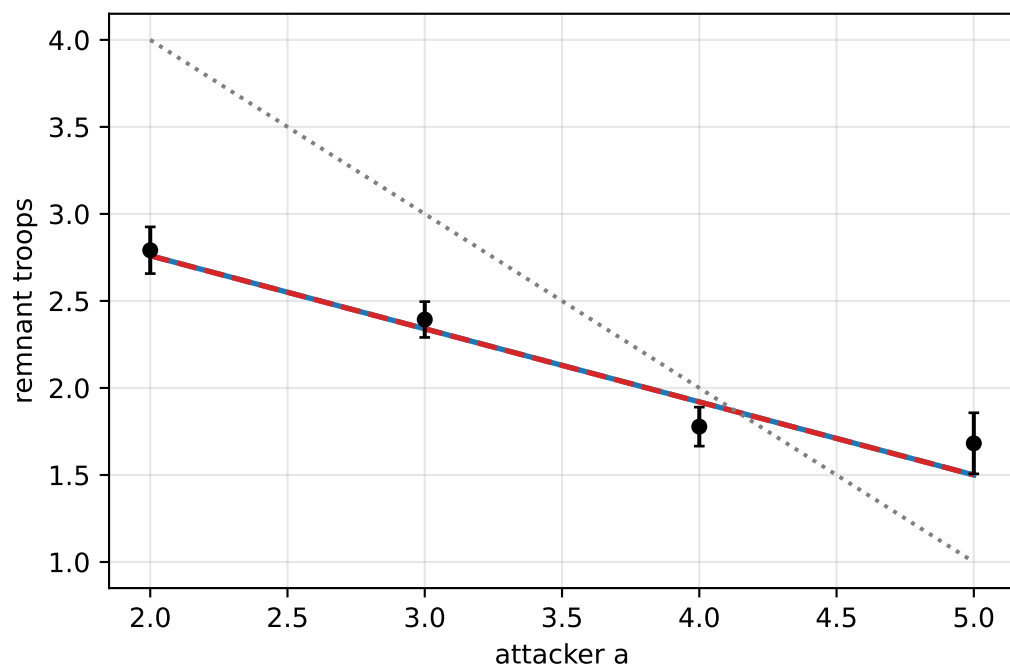
defender $d = 3$



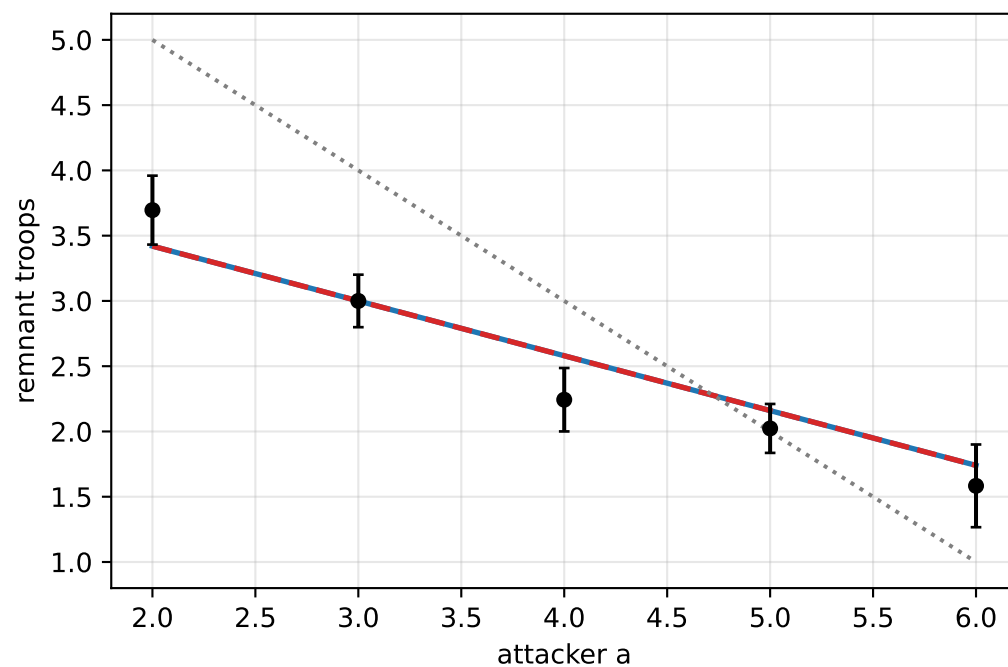
defender $d = 4$



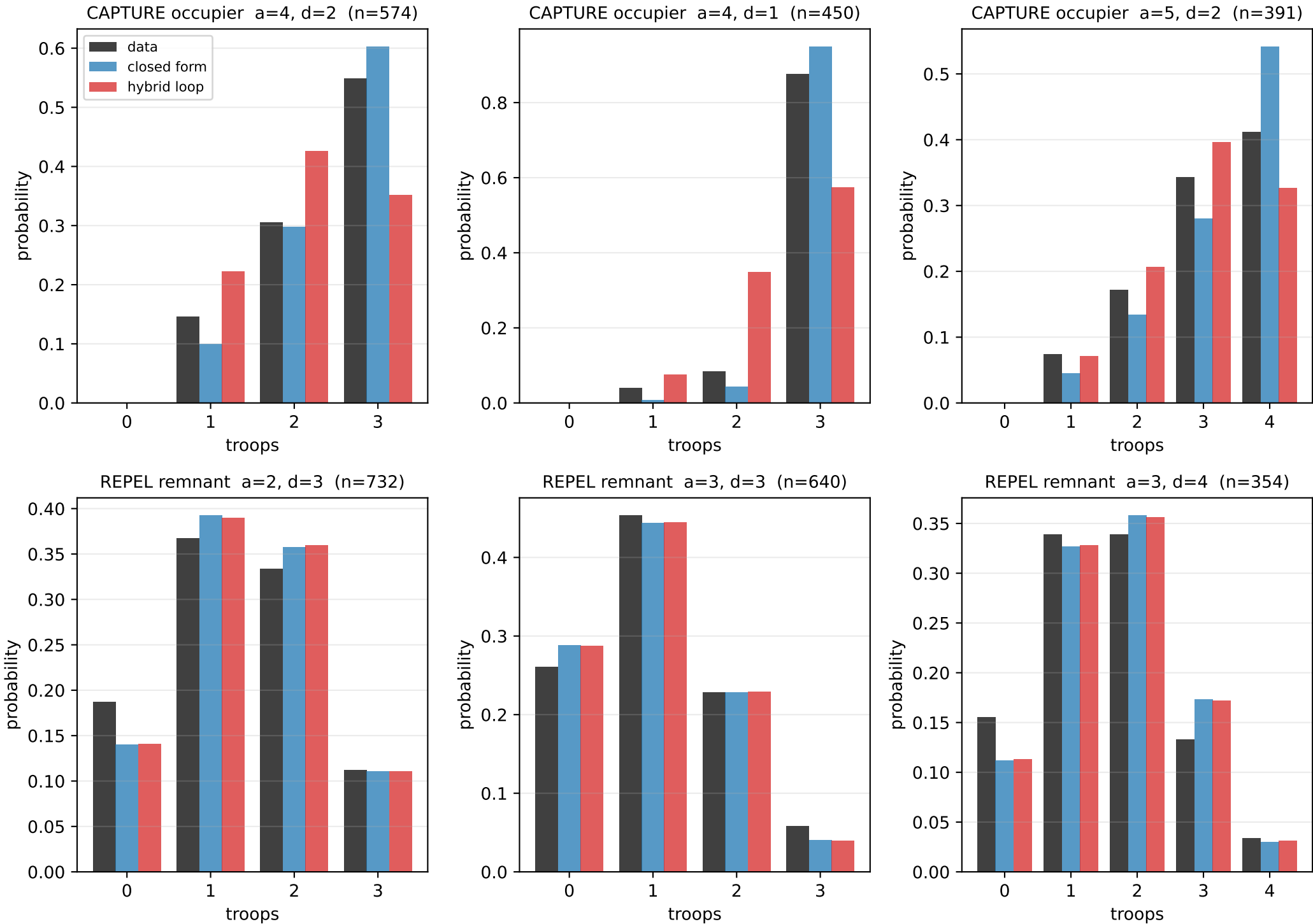
defender $d = 5$



defender $d = 6$



Survivor DISTRIBUTIONS — we match the spread, not just the mean



Per-cell calibration — predicted vs observed capture rate (point size $\propto$ #battles)

